

Online Appointment Scheduling System

Medical Clinic Scheduling

Term Project: Milestone 6 (M6)

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The section explains the methods used to monitor and control Clinic Appointment Scheduling operations which function in a simulated production environment. The system design establishes three main functions which include essential event logging, performance metric tracking, and failure management together with a health check endpoint to maintain system uptime.

Logging Plan

The system logs key events to help monitor and debug issues.

The following events are logged:

- User actions (viewing slots, booking appointments)
- Successful booking operations
- Failed booking attempts (e.g., slot already booked)
- System errors (exceptions, database failures)
- External service interactions (if applicable)

Example Log Messages

INFO:

- INFO: User accessed /slots endpoint
- INFO: Appointment booked successfully for slot_id=101

WARN:

- WARN: Attempt to book an already reserved slot_id=101

ERROR:

- ERROR: Failed to process booking due to database error
- ERROR: Unexpected exception in AppointmentController

Monitoring Metric

One key monitoring metric used in this system is:

- Number of bookings per hour

This metric helps track system usage and detect unusual activity.

Additional metrics:

- Failed booking attempts
- Average booking response time

Failure Scenario

A failure scenario occurs when two users attempt to book the same appointment slot at the same time.

One request succeeds, while the second request fails because the slot has already been reserved.

Recovery Strategy

The system handles this scenario by:

- Rejecting the second request
- Returning an error message to the user
- Logging the failed attempt

This ensures consistency and prevents double booking.

Health Check Endpoint

The system provides a health check endpoint:

GET /health

Example Response:

System is UP

This endpoint allows monitoring tools to verify that the application is running correctly.

Appendix (Code Snippets)

1. Logging Example:

```
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

private static final Logger logger = LoggerFactory.getLogger(
    AppointmentController.class);

logger.info("Booking request received");
logger.warn("Slot already booked");
logger.error("Database error occurred");
```

2. Health Endpoint Example:

```
@GetMapping("/health")
@ResponseBody
public String Health() {
    return "System is UP";
}
```

Conclusion

The Clinic Appointment Scheduling System receives complete monitoring and maintenance capabilities through this system management and logging plan. The system achieves dependable performance through its ability to monitor essential events and record vital statistics while using established methods to resolve system breakdowns. The health check endpoint enables instant system status checks which enhance system availability. The design supports better system visibility which makes it easier to identify and fix problems while enabling the system to function at greater capacity.